

Lab Records

Stack

Question 1:

Implement a stack using an array. Perform the following operations:

- PUSH 10, 20, 30
 - POP once
 - Display the final stack contents.
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Question 2:

Write a program to check whether a given string of parentheses is balanced using a stack.

Example:

Input $\rightarrow$ { [() ()] }

Output $\rightarrow$ Balanced

Question 3:

Convert an infix expression to postfix using stack.

Example:

Input $\rightarrow$ (A+B) *C

Output $\rightarrow$ AB+C*

Question 4:

Reverse a string using a stack.

Input: "HELLO"

Output: "OLLEH"

Question 5: Tower of Hanoi

Write a recursive program to solve the **Tower of Hanoi** problem for n disks.

Display the steps required to move all disks from **Source (A)** to **Destination (C)** using **Auxiliary (B)** rod.

Example:

Input $\rightarrow$ $n = 3$

Output $\rightarrow$

```
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from C to B
Move disk 3 from A to C
Move disk 1 from B to A
```

Move disk 2 from B to C
Move disk 1 from A to C

Queue

Question 6:

Implement a queue using a linked list.

Question 7:

Implement a **circular queue** using an array.

Perform these operations and show results:

`enqueue(10), enqueue(20), enqueue(30), dequeue(), enqueue(40).`

Question 8:

Reverse the first k elements of a queue using stack.

Example:

Input $\rightarrow$ Queue: [1, 2, 3, 4, 5], $k = 3$

Output $\rightarrow$ [3, 2, 1, 4, 5]

Question 9:

Trace the operations on a circular queue of size 5:

`enqueue(10), enqueue(20), enqueue(30), enqueue(40), dequeue(), enqueue(50), enqueue(60)`

Show final `front`, `rear`, and queue contents.

Question 10:

Write a program to check whether a given sequence of operations can be possible for a queue of given size.

Example:

Input $\rightarrow$ `enqueue, enqueue, dequeue, enqueue, dequeue`

Output $\rightarrow$ Valid sequence.
